

The Trackunit Way

The Way We Do Things Around Here



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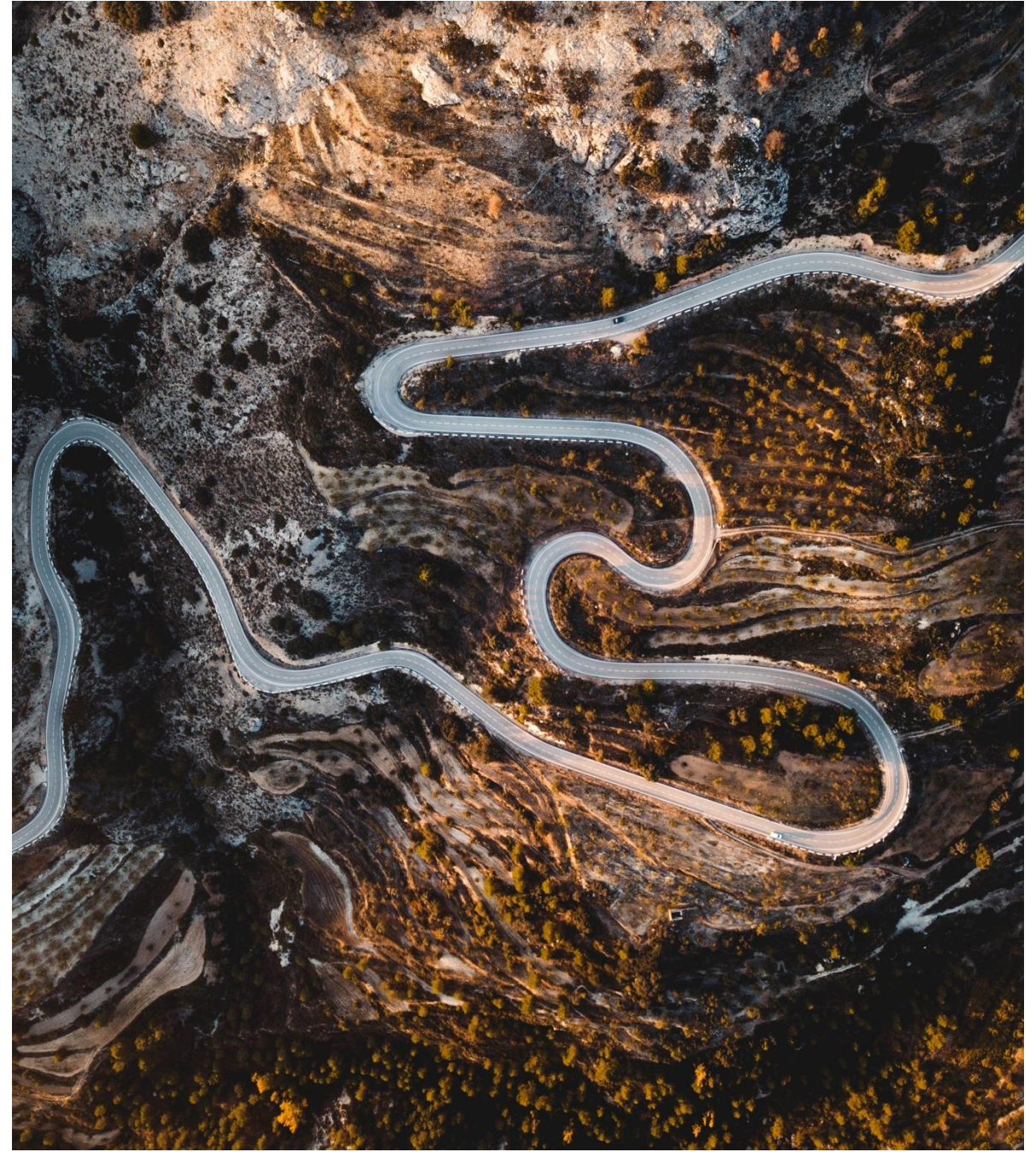
Staff Software Engineer





Agenda

- 01** **The Trackunit Way – Why we exist**
Purpose and Ambition
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Scaling Product Engineering within a SaaS company
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The Trackunit Way - Always in progress

The Trackunit Way is always in the making. It is a living, evolving mechanism that we want to work with in an intentional and deliberate way – what we do shapes who we are – and who we aspire to become.

We consider The Trackunit Way as a competitive advantage. While the world around us has changed dramatically over the past century, many of the ways organizations work - and think about people - have remained surprisingly unchanged. At Trackunit, we believe the way we work matters. And it has never been more important to be intentional about how we collaborate, lead, and grow together if we want to stay relevant, human, and competitive.

This is not an attempt to grasp the unique culture of Trackunit in all details. Neither is it meant as a step-by-step guidebook that is set in stone. It's made to spark inspiration and new ideas for those of us who've witnessed our evolutions, for new hires who will shape our future, and people considering to join us on our journey.

Welcome to our human operating system.





01

Purpose, and Ambition

Why We Exist



Be part of something bigger

At Trackunit, we're committed to construction - one of the largest industries in the world.

It's an industry of strong, hard-working people, building the infrastructure of our lives. An industry that used to be known for the cities it built and the landscapes it formed. For complex infrastructures and the rise of skyscrapers in unthinkable shapes and sizes.

Construction enables each one of us to live better lives through safer roads and rails, modern hospitals and schools, offices and homes. But for too long, the industry has been known for blown-up budgets and schedules delayed.

At Trackunit, we believe we can still do the impossible and rebuild the reputation of the industry.

New technology has emerged, and today we

know more than ever before. We build the services that enable construction to work smarter together and grow faster together.

An industry that has long been the laggard of digital transformation is now at a tipping point. Change is happening.

What impact are you going to make?

Facts & figures

- Circa 15 Trillion-dollar industry
- 6-8% of the global workforce works in construction
- Low single digit profit margins
- +20M pieces of construction equipment globally

Our purpose

Our purpose decoded

Our purpose statement decodes what we do. Each word is deliberately chosen and carries meaning. Purpose is personal - it connects to each of us in different ways. Use this as a guide to understand how your work contributes to our shared mission.

Our approach is human, collaborative and based on co-creation. We are driven by a design thinking approach, being empathetic, iterative and always striving for an ecosystem-wide impact.

Downtime is the core of all problems in the construction industry. We address it through five key areas, looking at downtime through the lens of machines, humans, companies, our industry as a whole and society at large. Eliminating Downtime is the contribution made to create an impact beyond the industry because it exists in the world, for the world.

Together, we Eliminate Downtime to build the most useful industry for the world

We are builders and doers by heart and committed to turn every relevant idea into a catalyst for change.

Being useful is the core DNA of our company. We always strive to walk on two legs, driving a highly commercial and impact-focused agenda at the same time.

Construction is a very big player in the world, and we have at least the same aspirations so that the industry can consciously create an impact for the world. By doing so, we deliberately balance an inwards focus with an outwards outlook because long-term relevance for any ambitious company will also be measured in the contribution outside of the industry.



How Do We Do It?

The construction industry is our home. A tough industry with cool machines and cool people. **We love machines and love to collaborate with people who love machines.**

We unite the construction industry to Eliminate Downtime through a unique human operating system and a superior data-driven platform - always with a highly commercial, sustainable, and global ambition.

We are a trusted ecosystem orchestrator making the largest data lake in the industry useful to Eliminate Downtime. This is our contribution to anyone who wants to deliver impact

today and tomorrow through insights, products, services, and business models – all developed by people and for people.

We operate as global team, compassionately challenging the dogmas in how to be an inspiring and relevant industry always co-creating for a real impact.



Nobody is better than everybody.

We trust each other and do the best of our work when we co-create and collaborate.



02

Past, Present, Future

Who We Are and Strive to Become



Our Journey

Trackunit started as a technology company pushing innovation and pioneering technological progress before IoT was a thing.

Since then, our journey has been a story of growth, learning, and doing the impossible. Trackunit is focused on the construction industry where we are delivering both hardware and software to the global market with an ambition to eliminate downtime and turn construction into the most useful industry for the world. The pioneering spirit is still intact – we drive for progress and impact, and the people are the true secret sauce to the journey.

How do we unlock creativity, empower and inspire people, and shape what's next for our industry?

That's what we're asking ourselves every single day in our pursuit to foster a workplace where we can all bring out our full potential and play to each other's strengths.

The journey ahead with scaling and building a platform for construction equipment calls for courageous ownership, compassionate collaboration, and a relentless focus on solving what truly matters, both now and in the long run.

Empowering people to act with clarity and purpose unlocks imagination, accelerates progress, and ensures we deliver on our collective promise.

Growing Our Footprint

Trackunit is the global market leader in brand-agnostic IoT solutions for construction equipment

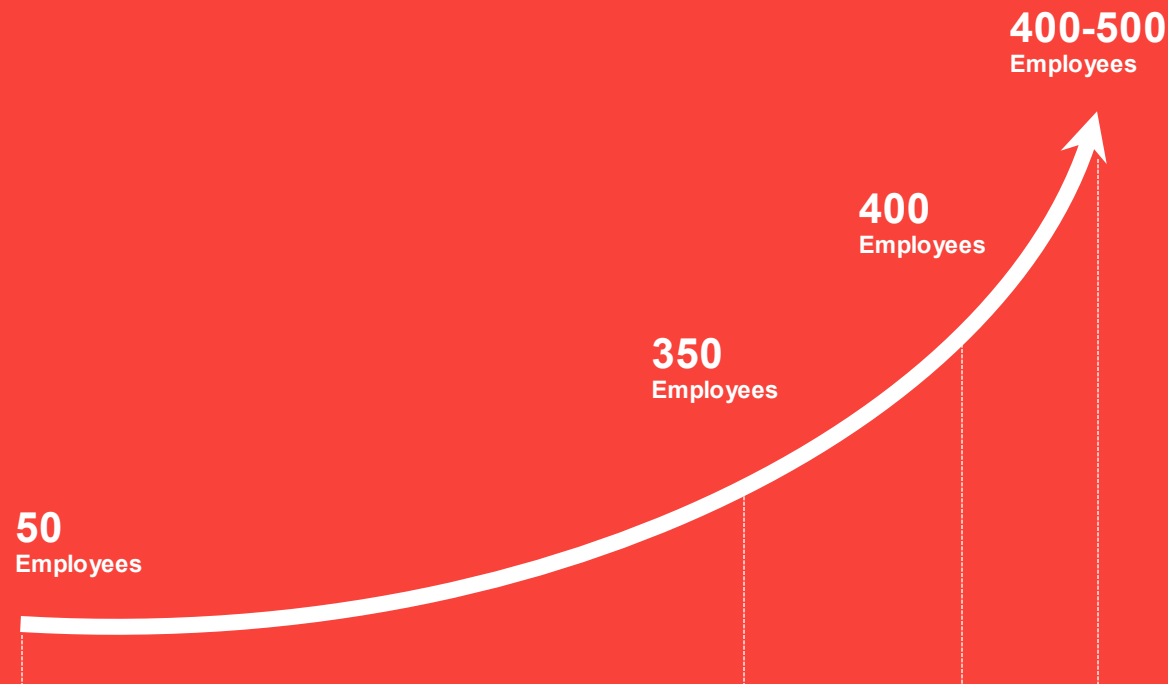
Half a decade ago, Trackunit was a local business providing hardware to track construction equipment.

Today, we have grown into a SaaS-company and a thought leader servicing the needs of the entire ecosystem of stakeholders within construction.

Doubling down on equipment

connectivity in existing markets and growing our network, Trackunit is ready to take the industry into a digital future.

Looking further into the future of the industry, Trackunit is well-positioned to facilitate and capitalize on the immense digitalization the construction industry is leaning into.



1998

M-Tec A/S established.

2003-15

Trackunit brand established.

Trackunit subsidiaries established across Western Europe.

Trackunit acquired by investment group of Goldman Sachs and GRO.

2016-21

Entry into the North American market.

Trackunit Iris platform launched to market.

Acquires Dreyer & Trimm, and Satrak.

Entry into APJ market.

Trackunit acquired by Hg Capital.

Trackunit acquires Industrial IoT division of ZTR.

2022

Circa DKK 1 billion net revenue.

Launches 5G, second-generation Spot connecting high-value, non-powered assets to ecosystem.

Stages first-ever Trackunit NEXT event.

Trackunit Kin wins Rental Awards Editor's Choice 2022.

2023

Posts net revenue of DKK 1.2b.

Acquires German ConTech Flexcavo, US deployment specialist OEMSi.

Launches Emissions Reporting solutions; new-gen Access Management.

Initial launch of Marketplace and Platform SDK to facilitate ecosystem co-creation and empower partners.

2024

More than 3 million assets connected.

Launches IrisX, the first operating data platform specifically tailored to the construction industry.

Introduces Trackunit Pass, new-gen Raw, Specialty Equipment, and Network Solutions to market.

Consolidates market-leading, thought leadership position on sustainability with publication of 'Constructing a better future' paper.



A global SaaS business

450+

People all over the world

13

Hubs all over the world

8.3/10

Employee Engagement

45+

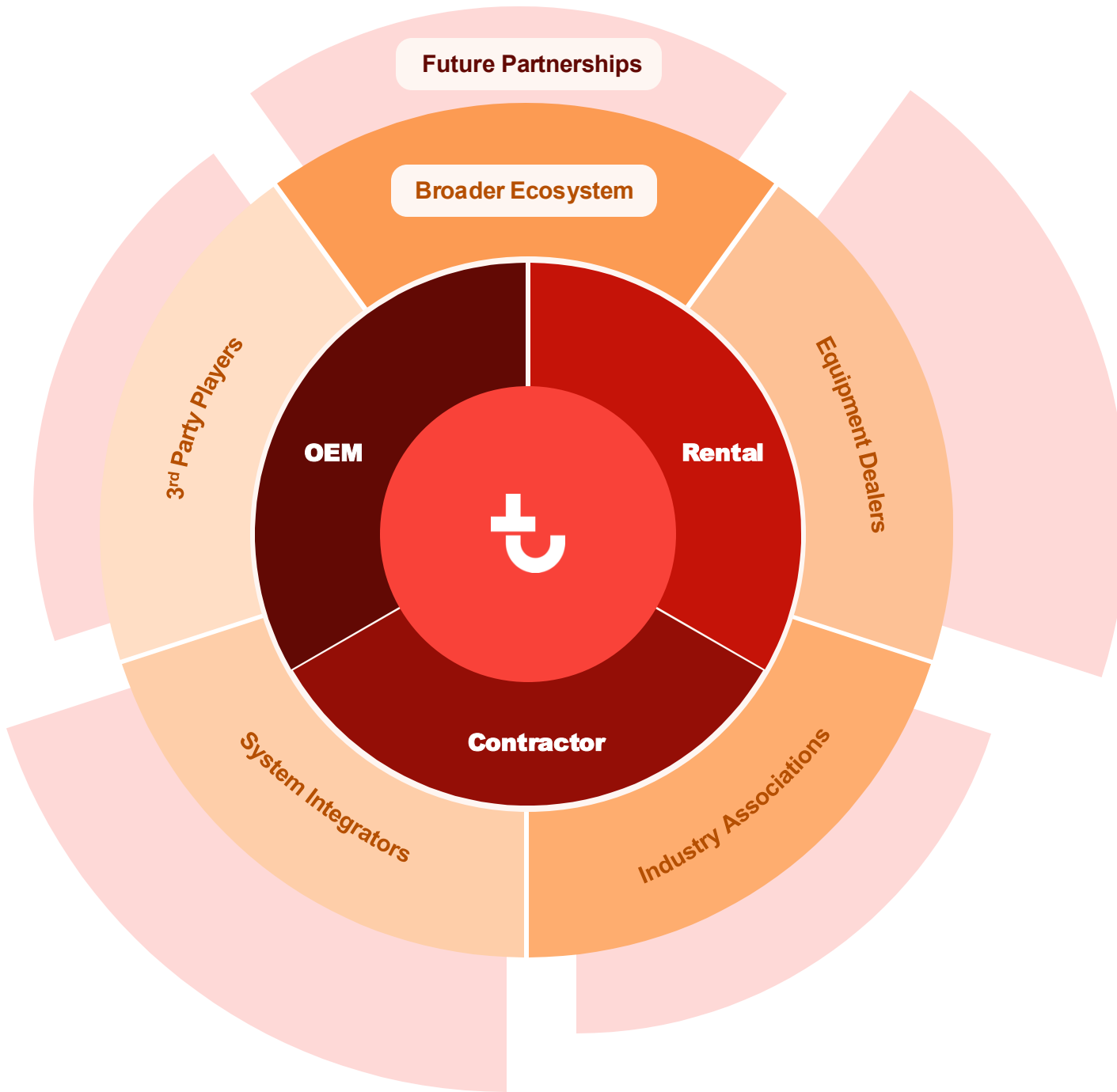
Nationalities



03

Our human operating
system

How We Deliver



Stewards of A Global Ecosystem

Connected by Purpose, Powered by Insights

The industry thrives on collaboration. At Trackunit, we bring industry professionals, partners, customers, and employees together to share knowledge, solve challenges, and drive meaningful change.

We operate within a **Connected Ecosystem**—an industry-wide network where people, machines, and insights come together to eliminate downtime.

By integrating data-driven insights with strong partnerships, we empower our ecosystem to work smarter, safer, and more efficiently.

Our role? We are more than a technology provider; we are an **ecosystem orchestrator**, ensuring that every connection—whether digital, human, or operational—creates value.



A Human-Centric Organization

At Trackunit, we don't have employees - we are people on a mission together.

Our success as a company is built on a deep understanding of human nature. We design our organization to reflect how people truly flourish - embedding trust, care, and purpose into everything we do.

Our ambition is clear: To set a new standard for what a truly human-centric organization looks like.

We Work with Purpose

Everyone at Trackunit should have the opportunity to do meaningful work. We help people connect their strengths to our mission - because when work matters, impact follows.

We are Built on Trust

We trust each other to take ownership, make an impact, and shape the future of Trackunit. We don't micromanage - we empower.

We Make Life Work

Work should be a fulfilling part of life, not something that competes with it. We support flexibility and balance, so people can bring their best selves to work and beyond.

We Unfold Human Potential

We create an environment where people do their best work and grow with us. We unlock potential by being close enough to see it, and by creating opportunity, and support to unfold it.

We Design for Belonging

One size fits no one. We see and value each person's unique contributions appreciating that complementarity is what make our teams perform at their best.

We Win Together

Nobody is better than everybody. We succeed when we collaborate, challenge each other, and build something bigger than ourselves.



Orchestrating a Community

It takes a village to raise a child they say.

A big, wicked challenge like eliminating downtime is also something that calls for an entire community to come together. Research shows that people flourish and work at their best in organizations that establish meaning, autonomy, freedom and mastery.

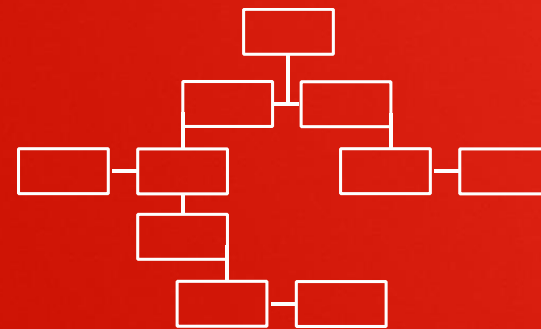
That's why we do everything we can to keep our hearts, minds, and brains open – just like we strive to keep our organizational boundaries open.

To be something for the world, we're on a journey towards an unlimited organization and to orchestrate a sense of community across functions and levels of the industry. We work outside in – always co-creating with customers, partners, competitors, associations, freelancers, and thought leaders.

Our bias for building trust and connection is what sets us apart from the rest.

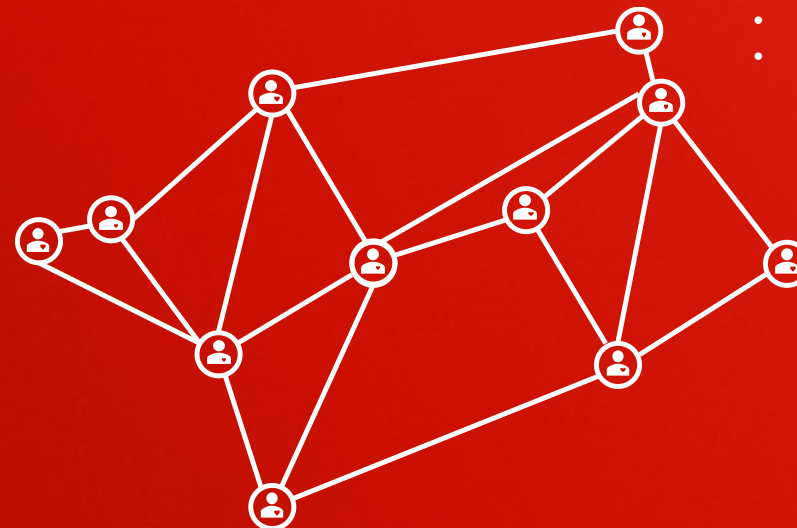
A Traditional organization

- Rules
- Process
- Policies



An Unlimited Organization

- Meaning
- Autonomy
- Freedom
- Mastery





The Principles We Live by

What we should know and do to deliver on our purpose and strategy.

Show Compassion.

We practice compassion by actively listening without judgment, seeking to understand before responding. We approach conversations with curiosity, self-awareness, and emotional intelligence, ensuring that feedback is delivered with honesty, respect, and a genuine desire to help. While empathy allows us to recognize others' emotions, compassion moves us to take action – supporting our colleagues in thriving and growing.

Win now. Win later.

We stay focused on our ultimate goals, knowing that if we fixate only on the next step, we risk losing our balance. To drive real impact and achieve our business objectives, we solve the problems that truly matter with curiosity, agility, and a commitment to learning. We move beyond emotional arguments, instead articulating and testing hypotheses with data, facts, and proof points. Winning isn't just about Trackunit - it's about catalyzing the ecosystem and creating lasting impact beyond our company.

Own Your Part.

We take responsibility for achieving real results by owning our part in every challenge and opportunity - while ensuring it is well-integrated with the bigger picture. We engage others, foster a strong sense of ownership, and follow through on our commitments. Organizational change starts with personal change, so we hold ourselves 100% accountable for our actions and words - and we support those around us in doing the same, recognizing that true ownership includes collaboration and alignment with others.



04

Scaling Product Engineering within a SaaS company

New Product Tracks



**“ Empowered
product teams
solving real
problems**





With Increased Focus

Alignment

Our strategy and purpose are key elements of New Tracks.

With alignment around purpose and strategy we want to create aligned empowerment.

Craftmanship

Working across multiple dimensions; design, usability, engineering and product management to solve real customer problems right

Together

With team autonomy, mastery & purpose at forefront we infuse deeper support through cross functional collaboration



The 4 key areas of product craftsmanship:

Customer Value

*Does the proposed solution provide real value to our customers?
Will they buy it or choose to use it?*

Business Viability

Will this solution work for the various dimensions of our business?

Usability

Can customers and users figure out how to use it?

Feasibility

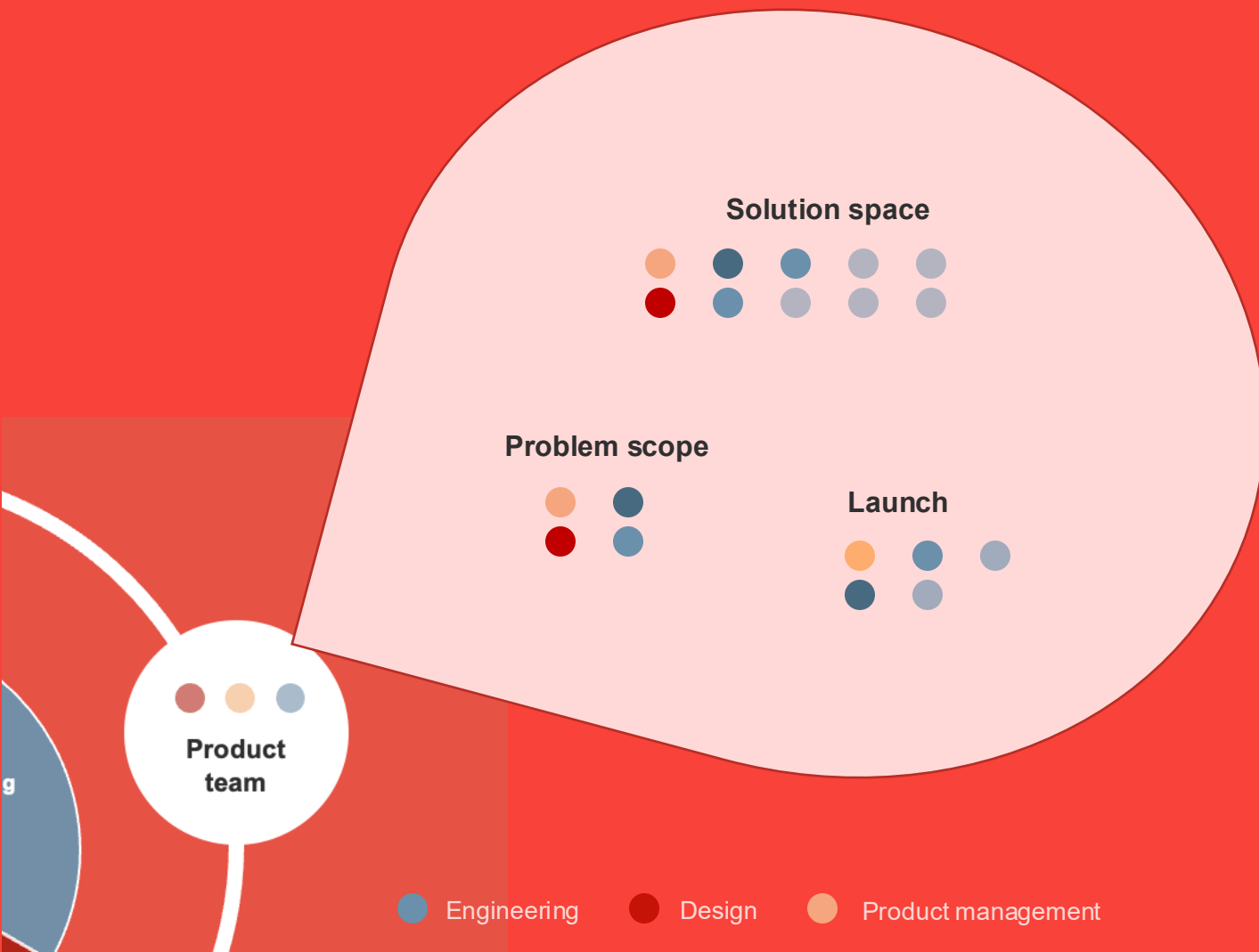
Can we build it with the time, skills, and technology we have?

A **product manager** is accountable for ensuring **value** and **viability**

A **design lead** is accountable for ensuring **usability**

An **engineering lead** is accountable for ensuring **feasibility**

However, the team members have a **shared responsibility** for this and **truly collaborate** in an intense, give and take, to discover the best solution for the market.



Product teams

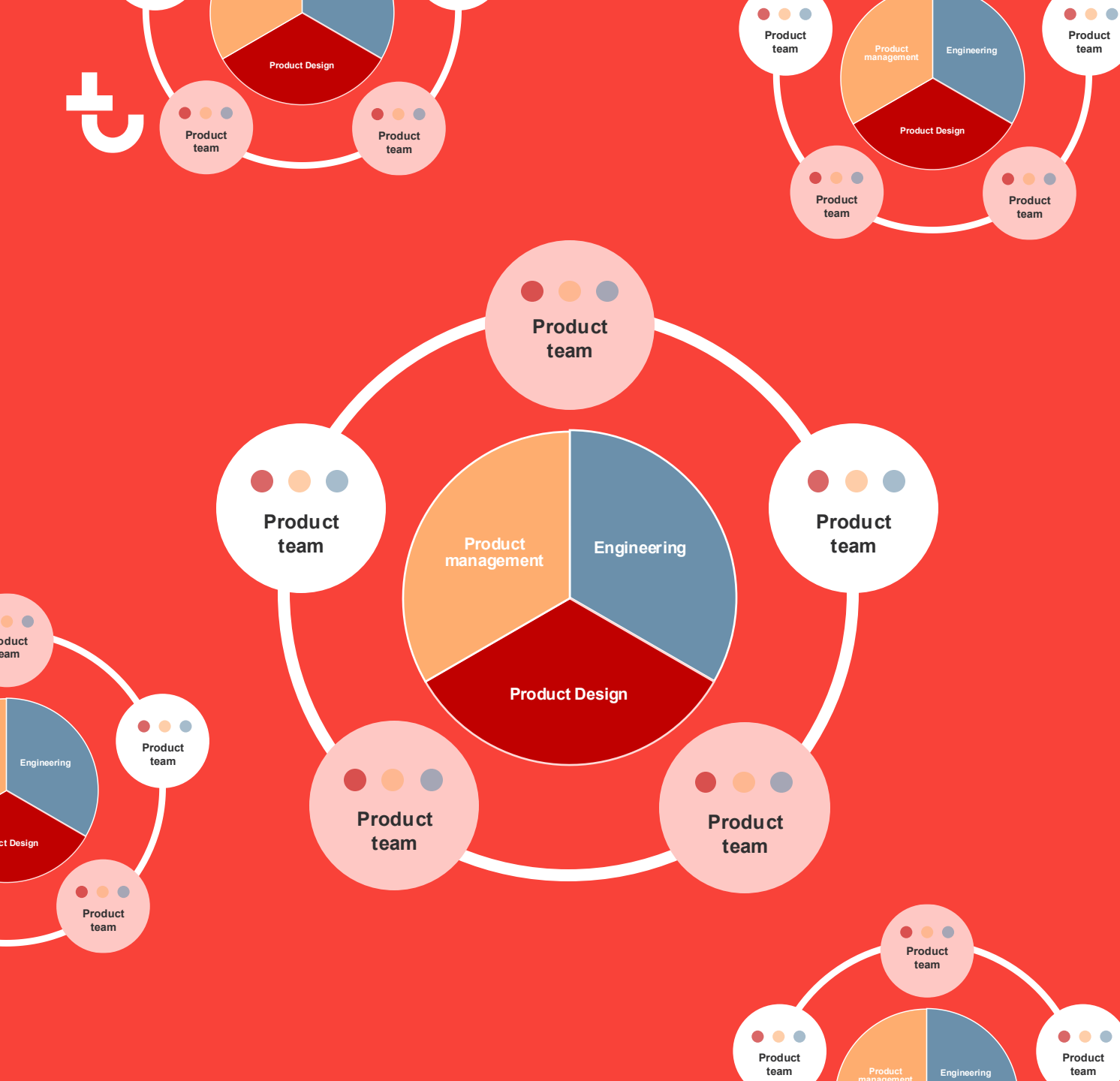
With **New Product Tracks** we changing from Feature teams to Product teams – increased focus on business outcome.

Staffing to match the process stage.

Product teams are empowered, autonomous and accountable for defined problems and solutions to these.

The portfolio lens adds the following:

- Work towards the portfolio objective
- Align on strategy across teams
- Increased involvement in decisions across teams
- Support from leads in alignment, strategy and craftsmanship.



Portfolio teams

With **New Product Tracks** we are introducing product portfolios to better support our product teams

A Portfolio org is 2-5 product teams under shared leadership.

Portfolio leads are aligned across product management, engineering and design. Their job is to support the teams with:

- Strategic alignment
- Distributed decision making
- Product craftsmanship



How are we organized?

<https://trackunit.atlassian.net/wiki/spaces/Products/pages/4823908420/Overview+Product+Areas>



Product Roadmap

<https://trackunit.com/product-portal/>

Trackunit Product Portal

Your invitation to shape the future of our solutions

The Trackunit Product Portal is the window to see the solutions we are planning, building, and have launched in the Trackunit platform. The Portal is an invitation to our customers, partners, and the construction industry to actively shape the future of our products.

Providing this view into the future roadmap sets a new standard for industry transparency to accelerate collaboration on our industry-wide mission to Eliminate Downtime.

The screenshot displays the Trackunit Product Portal interface. At the top, there are navigation tabs: LAUNCHED, EARLY ACCESS, BUILDING, PLANNING, and OPPORTUNITY. The main content area is divided into two sections: H2 2025 and H1 2025. Each section contains several cards representing different product features or updates. The H2 2025 section includes 'Attention List' (with a magnifying glass icon and a count of 1), 'Self-serve and get agent support from Trackunit...' (with a magnifying glass icon and a count of 0), and 'SELF-SERVE AGENT SUPPORT' (with a magnifying glass icon and a count of 0). The H1 2025 section includes 'Service set up for retrofit assets' (with a magnifying glass icon and a count of 0), 'Device Administration App' (with a magnifying glass icon and a count of 0), 'Operating Modes' (with a magnifying glass icon and a count of 0), 'Asset Notes' (with a magnifying glass icon and a count of 2), 'Global Search in Trackunit Manager' (with a magnifying glass icon and a count of 1), 'Show & Hide Assets' (with a magnifying glass icon and a count of 0), 'Fleet Map Geographical Search' (with a magnifying glass icon and a count of 0), and 'Display past movements of an asset' (with a magnifying glass icon and a count of 1). Each card features a red background with white text and icons, and a small magnifying glass icon with a count next to it.



Product Engineering Demoes - example

2030 STRATEGIC VALUE DRIVERS

- | | | | | | |
|---|---|---|---|---|--|
| 1 | Scale core connectivity base across customer segments and geos (incl. APJ) | 2 | Expand and reinforce value proposition to customers via new Value-Added Services | | |
| 3 | Leverage data lake to create unrivalled insights via AI & ML | 4 | Facilitate engagement between customers to drive network effects | | |
| 5 | Enable the connected site ecosystem for construction | 6 | Unlock new capabilities at the edge via device and SW innovation | 7 | Unlock ecosystem value via the platform and marketplace |

PURPOSE

(What customer problem are we solving?)

- Enabling customers to either self-serve or automate their processes
- Remove customer dependency on Support – freeing time on our side for other requests and removing unnecessary wait times on theirs

ROADMAP

Latest

Launched and in maintenance mode

Next

Re-engagement efforts

Next+1

New data sharing mechanism through CPs

SDLC PHASE

Experiment
[Stage 1]

Design
[Stage 3]

Validate
[Stage 5]

Maintain
[Stage 7]

Define
[Stage 2]

Build
[Stage 4]

Release
[Stage 6]

EOL
[Stage 8]

STATUS/WHAT'S NEW?

(What are we demoing today?)

- Taking a walk down memory lane
- Current process – prerequisites, options, Manager functionality



Product Development Lifecycle (PDLC)

Overview

- **4 Core Phases**

Customer-Problem Fit → Problem-Solution Fit → Product-Market Fit → End of Life.

Ensures customer needs are understood, validated, solved, and scaled sustainably

- **8 Stages**

Define the “how” of delivery, guiding execution and tangible outcomes.

- **Gates (Decision Checkpoints)**

Formal review points between stages for go/no-go decisions.

Ensure accountability, alignment, and quality before progressing.

- **Governance Model (DRCI Framework)**

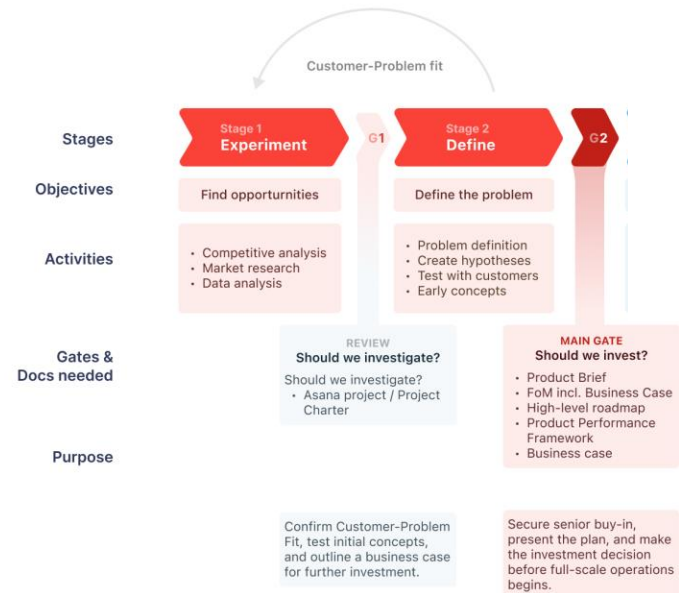
Driver, Responsible, Consulted, Informed roles clarify ownership.

Promotes transparency, efficient collaboration, and clear decision-making.



PDLC – Detailed view

- MAIN GATE** SLT decision point for go/no-go on investment, launch, or deprecation.
- REVIEW** Leader-level review to validate progress and log accountability.





SDLC

**Team creates design doc
describing
solution and implementation**



**Architecture group reviews
design doc**



**Team implements in an
iterative fashion**



COMPONENT — SERVICE

Design-Docs ☆

Owner
engineering-architecture-group

Lifecycle
production



OVERVIEW

DOCS

DEPENDENCIES

CIRCLECI

PULL REQUESTS

KUBERNETES

SNYK

SENTRY

API



[Asset MetadataScore
Release1 202203](#)
AssetApiEditBasicMetadata
202209
Assets ExposeRelayedBy
Release1 202209
Assets
MachineShareTransfer
Release1 20220801
Auditlog
GeneralAvailability
release1 202301
Authorization
MobilePermissions
Release1
Authorization
PersistedData Release1
CustomFields release1
202201
Deep Linking In Manager
202210
Digital Key 202211
Events PlaceBasedAlerts
Release1 202212
FileStore API Release1
202203
Integrations
IsoFeedScheduler
Release1 202212
KinGateway
PinnedLocation release1
202206

Asset Metadata Score



- Start Date: 2022.02.14
- Design Doc PR: [here](#)

Context & Scope

- A summary of the project/feature/design decision. Please describe it in the context of the landscape in which it's being built.
- Near term goal: help our customers improve quality of their assets' metadata
- Long term goal: more complete metadata will provide better insight and analytics capabilities
- Estimated size of this feature: **S/M**

Goals & Non-goals

- Technical goals: - provide metadata score for non-machine assets - align calculation of non-machine and machine assets' metadata score - include non-machine assets in fleet average metadata score calculation
- Non-goals: - users can edit only metadata of the assets they own, editing metadata and thus impacting the metadata score of visible, not editable assets is out of the scope
- Value for customers: - increased awareness of completeness of their assets' metadata
- Aimed for all customers using the new Manager for management of assets of their fleets
- No Mock-ups
- Input fields included in MetadataScore: Type, Production Year, VIN/Serial Number, Model, Brand. Each field has weight of 20%. After filling all of them, the score will be 100%.

Table of contents
Context & Scope
Goals & Non-goals
Proposed design
APIs
Data Storage
Cross Cutting Concerns

← Previous
2021 12 20 iso feeds self service

Proposed design

Next
AssetApiEditBasicMetadataForMachines 202209 →

SDLC - Design docs

When to use design docs:

- Make a substantial change to existing TU components and/or Features
- Introduce new features to the TU platform
- Add a new K8S deployment
- Use a new technology not on the Tech Radar

You do not need to go through the design review process for - Bug fixes - Smaller changes to existing components, that doesn't affect customers

Submitting a Design Doc

- Go to Design Docs repository
- Create a branch
- Copy the `template.md` Design Doc template, naming it `domain-feature-<release1>-yyyyymm.md`
- Place it in the `/docs` folder
- All dependent assets of the design doc should be placed in it's own folder `domain-feature-<release1>` + In case of diagrams they must be in a format viewable in githubs editor, e.g. png + In case the viewable format is created from a source this source must be included, e.g. drawio.xml
- If multiple releases/drops are not anticipated the "release1" part of the naming is optional
- Submit a pull request
- Trackunit/engineering-architecture-group should be automatically assigned as reviewers to the PR



SDLC – team responsibilities

Responsibilities of an component owner

- **Observability:** Set up structured logging, metrics, and traces. Alert on known bad states. Create dashboards to provide an overview of the component if needed.
- **Data Backup:** Follow standard backup procedures or manage independently.
- **High Availability:** Ensure components can handle failures gracefully and remain operational.
- **Right-Sizing and Cost Management:** Optimize resource allocation to avoid unnecessary costs, especially in non-production environments, and maintain cost estimates.
- **Performance:** Deliver responsive and efficient components for optimal user experience.
- **Stress Testing:** Validate component resilience and stability under load.
- **Security Compliance:** Adhere strictly to security standards including OWASP Top 10 guidelines, authentication via Okta and JWT validation, role-based authorization, [Vulnerability patch management](#), secrets management in Vault, and cryptographic policy compliance.
- **Quality and Testing:** Follow coding standards and maintain comprehensive automated testing, including continuous integration and smoketests.
- **Documentation:** Document component design, purpose, operational details, and internal logic clearly.
- **OpsLevel Compliance:** Maintain at least bronze level compliance in OpsLevel.
- **Ownership:** Clearly define and regularly update ownership records.
- **Tech Radar Compliance:** Use approved technologies to streamline skills and ensure maintainability.
- **Scalability:** Design components to scale horizontally, accommodating future business growth.
- **Interruptibility:** Components must handle interruptions gracefully, avoiding downtime or data loss.
- **GitOps Deployment:** Use Git for deployment configurations to enable effective version control and rollback capabilities.
- **Metrics and Logs:** Provide necessary metrics for monitoring and maintain standardized logging practices.
- **Auditability:** Log significant user actions. [audit logged](#).
- **Public API First:** Dogfood our APIs internally and consistently follow our public API-first strategy. all public APIs should adhere to the [public API guidelines](#).



06

With high empowerment comes great responsibility

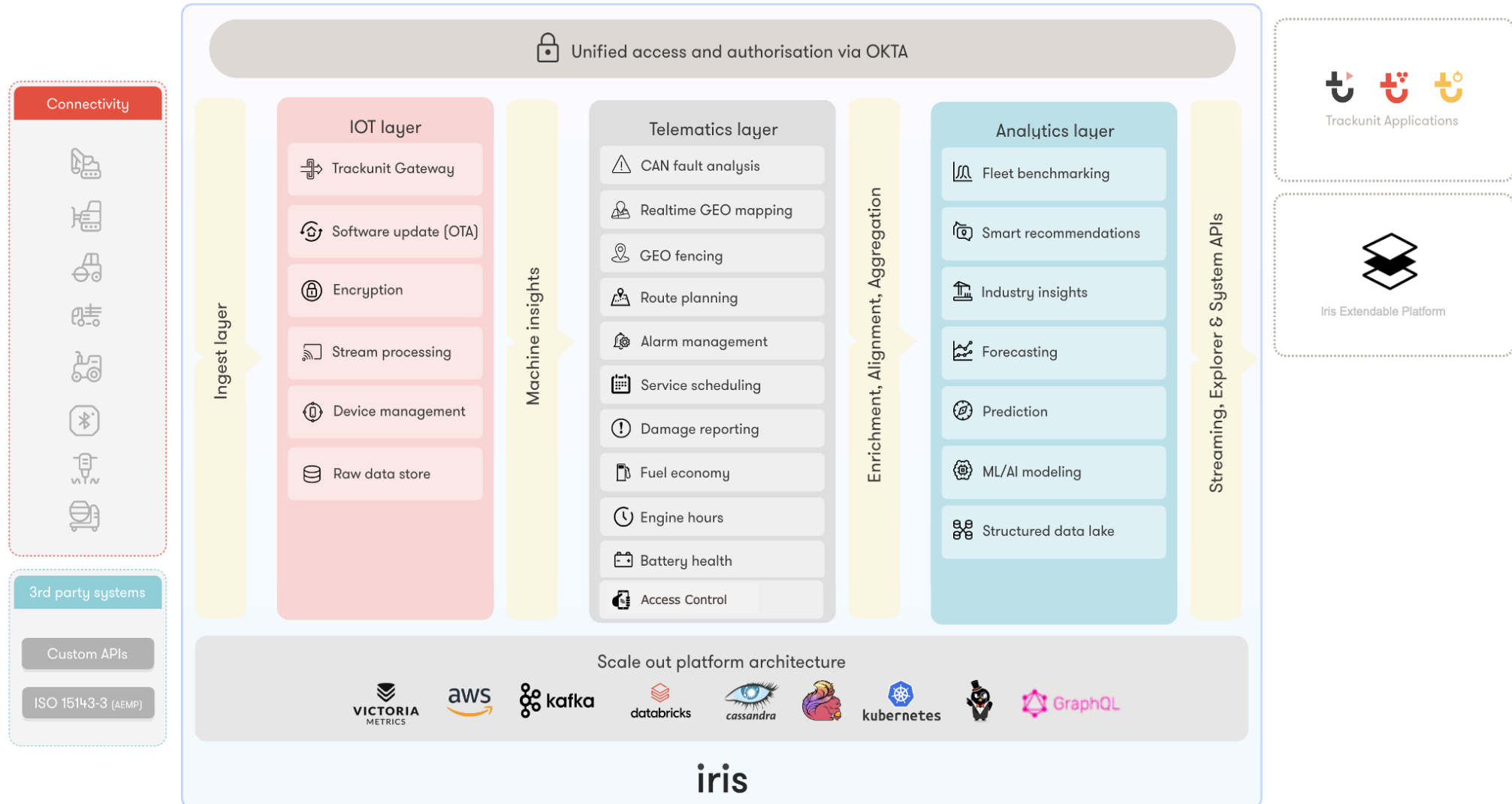
**You build it you
run it**



IRIS – Platform overview

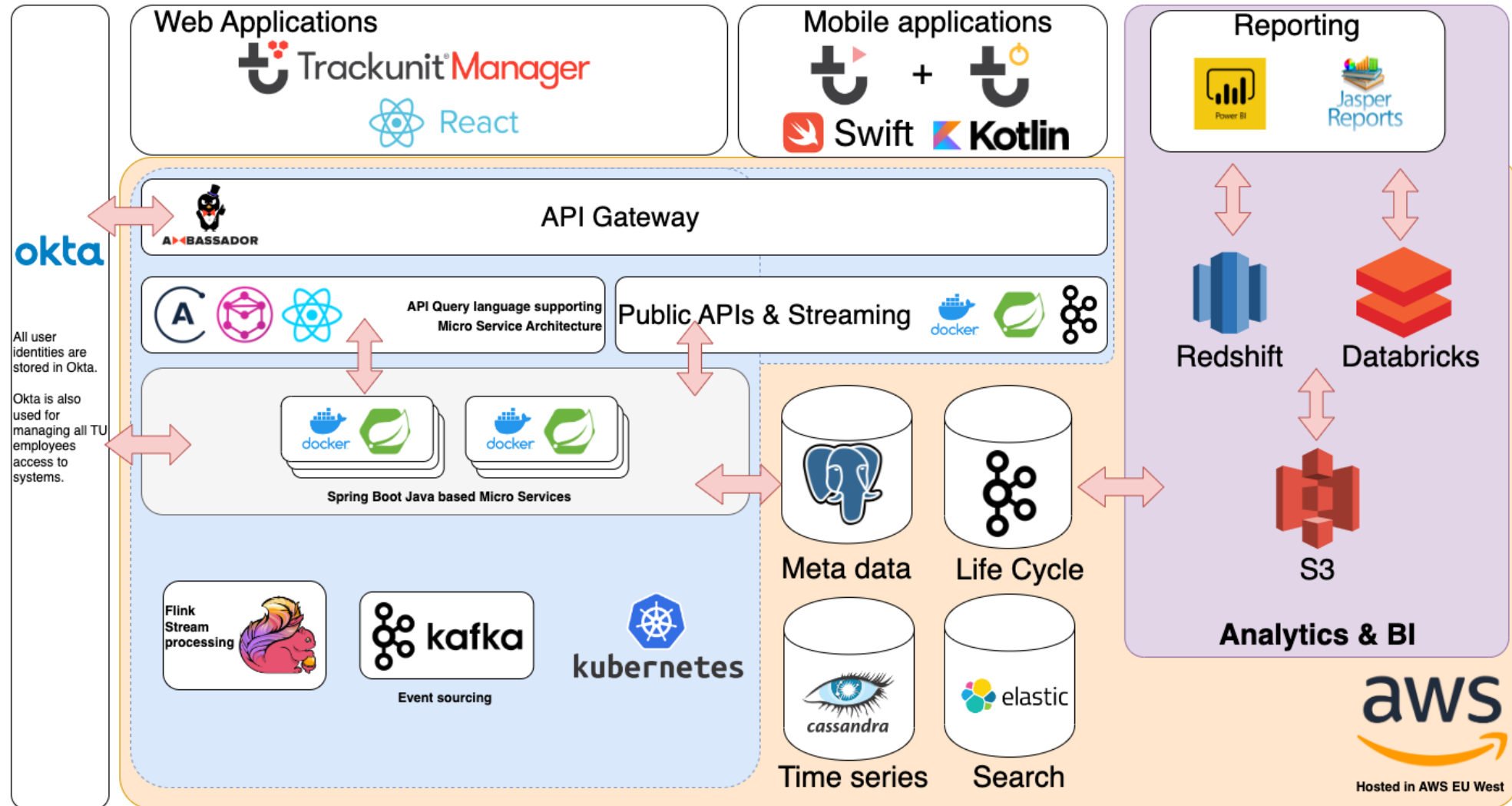


Iris architecture - Business perspective





Iris architecture





Iris in numbers

Flink jobs

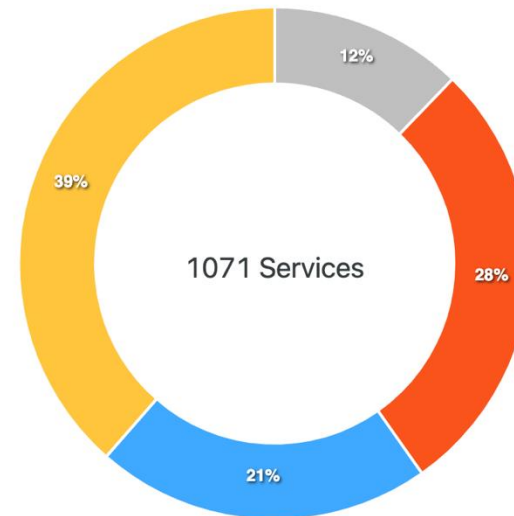
More than 100+ flink jobs

Created: 2022-10-11, 13:18:01	1.89G	Autopilot	Disabled	owner: team-splex	ⓘ	RL
Created: 2023-01-03, 13:07:18	2.99G	Autopilot	Disabled	owner: team_access_control	ⓘ	TRAN
Created: 2020-07-15, 14:00:02	1.91G	Autopilot	Disabled	owner: team_jo repository: Telematics2	ⓘ	RL
deploy Created: 2022-12-09, 15:05:38	2.94G	Autopilot	Disabled	owner: team_oem	ⓘ	RL
Created: 2020-07-16, 11:22:47	1.91G	Autopilot	Disabled	owner: team_jo repository: Telematics2	ⓘ	RL
Created: 2022-05-09, 16:20:22	1.91G	Autopilot	Disabled	owner: team_lot_data	ⓘ	RL
Created: 2020-07-16, 10:35:24	2.66G	Autopilot	Disabled	owner: team_jo repository: Telematics2	ⓘ	RL
Inventory-asset-detector Created: 2023-03-15, 13:28:52	4.83G	Autopilot	Disabled	owner: team_ai	ⓘ	RL

APIs and components

More than 130+ APIs and 600 frontend components

Non compliant (12%) Bronze (28%) Silver (21%) Gold (39%) ⋮



More than 1.9 mill lines of code analyzed daily

For security issues, code smells, bugs and other

ent PRIVATE

3, 5:01 AM • 3.6k Lines of Code • Java, XML

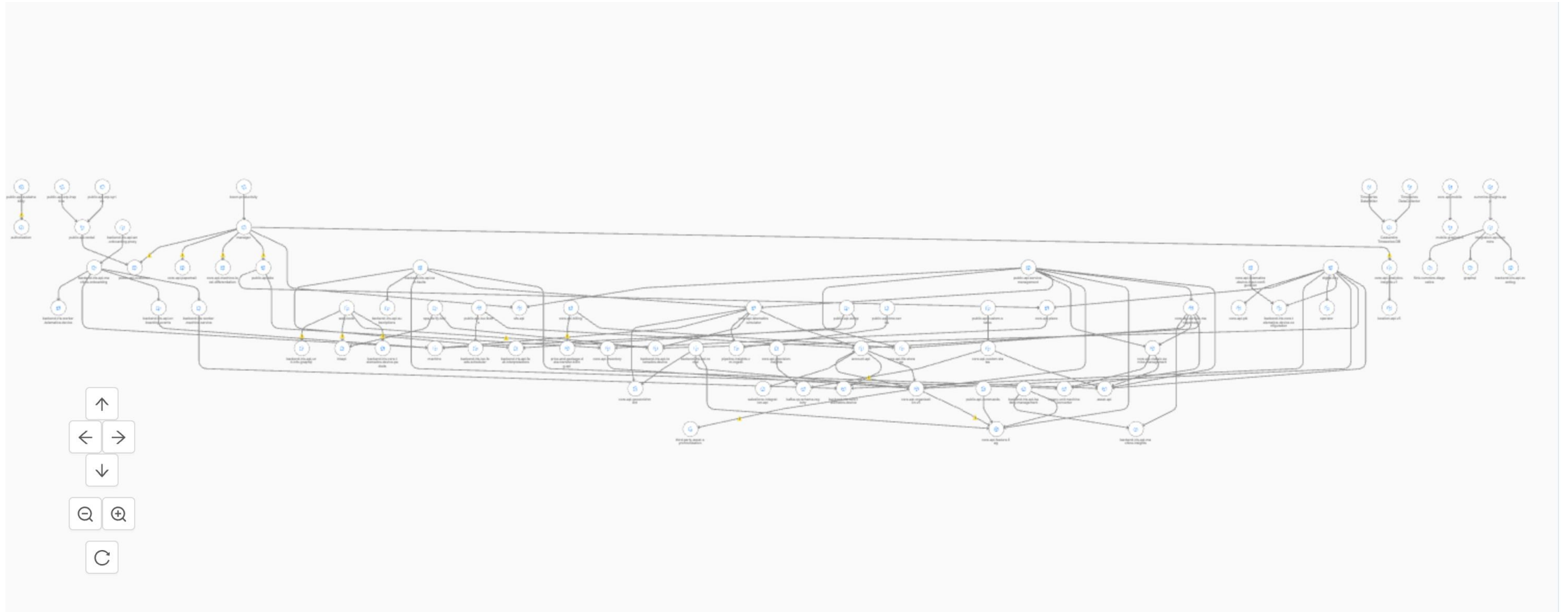
E 0.0% A 41 88.1%
es Hotspots Reviewed Code Smells Coverage

avro PRIVATE

ret. [Configure analysis](#)

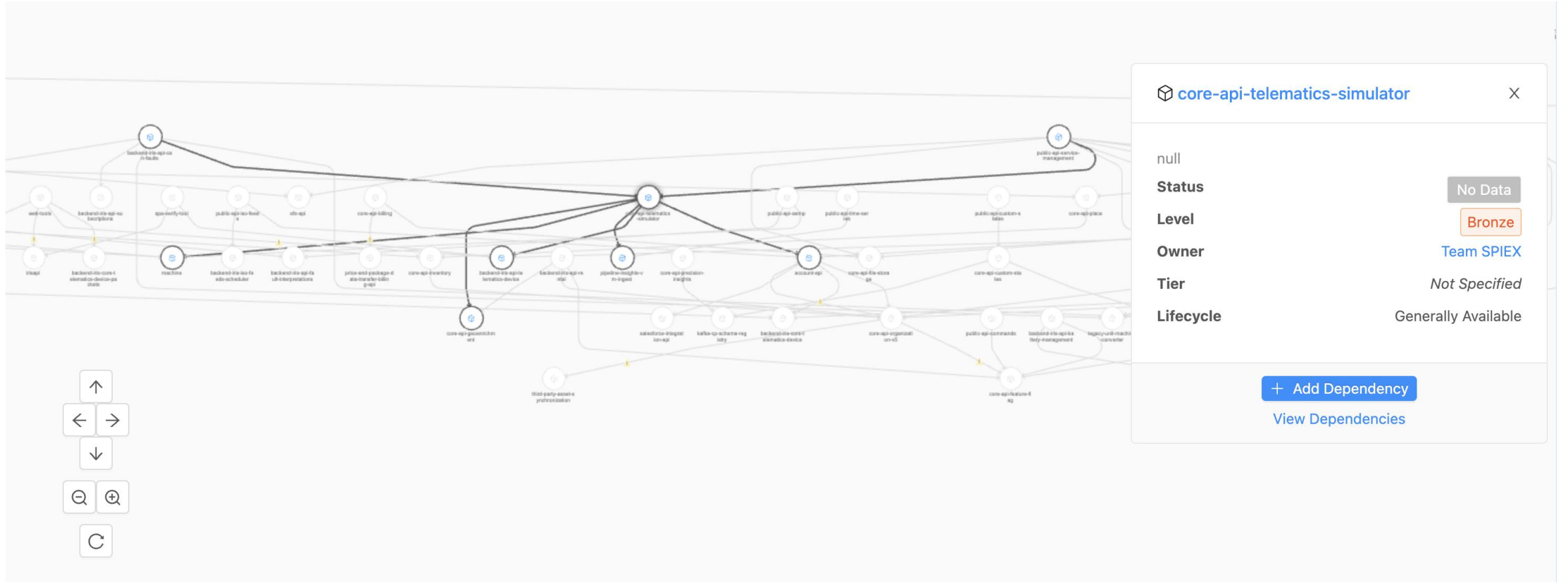


Interconnected microservices





Interconnected microservices



Trackunit Technology Radar Version #1

QUADRANT 1 ZOOM IN

Languages & Frameworks

A selection of programming languages, alongside essential frameworks for building a variety of custom software.

QUADRANT 2 ZOOM IN

Methods & Patterns

Key software development methods and design patterns, covering everything from continuous integration and testing to architecture.

QUADRANT 3 ZOOM IN

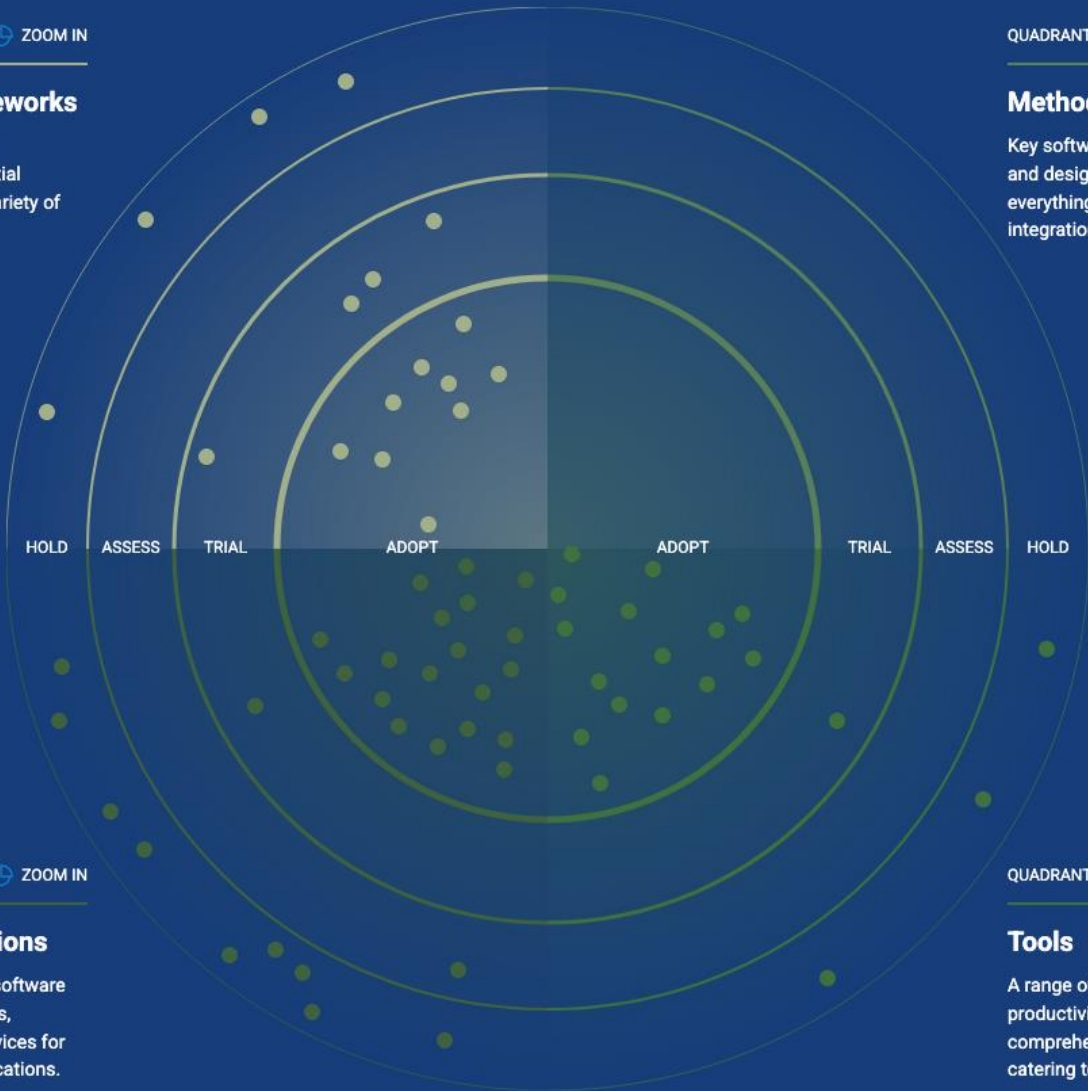
Platforms & Operations

Technologies and tools for software and infrastructure operations, including platforms and services for managing and scaling applications.

QUADRANT 4 ZOOM IN

Tools

A range of software tools, from simple productivity enhancers to comprehensive project solutions, catering to various project needs.



- ▲ New in this version
- ◆ Recently changed
- Unchanged

Filter by Tag

- ANALYTICS
- API
- FIRMWARE
- FRONTEND
- INFRASTRUCTURE
- MOBILE
- PROCESSING PIPELINES

Languages & Frameworks

ZOOM IN

ADOPT	TRIAL	HOLD
GraphQL	gRPC	Akka
Java		Play
Spring Boot		
Spring Data REST		
TestContainers		

Component ownership





SDLC – team responsibilities

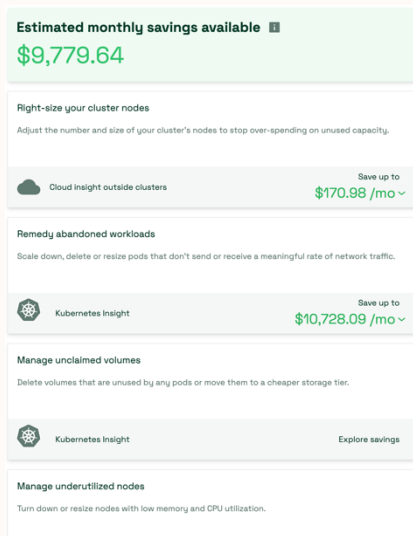
Responsibilities of an component owner

- **Observability:** Set up structured logging, metrics, and traces. Alert on known bad states. Create dashboards to provide an overview of the component if needed.
- **Data Backup:** Follow standard backup procedures or manage independently.
- **High Availability:** Ensure components can handle failures gracefully and remain operational.
- **Right-Sizing and Cost Management:** Optimize resource allocation to avoid unnecessary costs, especially in non-production environments, and maintain cost estimates.
- **Performance:** Deliver responsive and efficient components for optimal user experience.
- **Stress Testing:** Validate component resilience and stability under load.
- **Security Compliance:** Adhere strictly to security standards including OWASP Top 10 guidelines, authentication via Okta and JWT validation, role-based authorization, [Vulnerability patch management](#), secrets management in Vault, and cryptographic policy compliance.
- **Quality and Testing:** Follow coding standards and maintain comprehensive automated testing, including continuous integration and smoketests.
- **Documentation:** Document component design, purpose, operational details, and internal logic clearly.
- **OpsLevel Compliance:** Maintain at least bronze level compliance in OpsLevel.
- **Ownership:** Clearly define and regularly update ownership records.
- **Tech Radar Compliance:** Use approved technologies to streamline skills and ensure maintainability.
- **Scalability:** Design components to scale horizontally, accommodating future business growth.
- **Interruptibility:** Components must handle interruptions gracefully, avoiding downtime or data loss.
- **GitOps Deployment:** Use Git for deployment configurations to enable effective version control and rollback capabilities.
- **Metrics and Logs:** Provide necessary metrics for monitoring and maintain standardized logging practices.
- **Auditability:** Log significant user actions. [audit logged](#).
- **Public API First:** Dogfood our APIs internally and consistently follow our public API-first strategy. all public APIs should adhere to the [public API guidelines](#).

Component ownership

Cost

You are responsible for making informed decisions that lead to a reasonable cost of your services



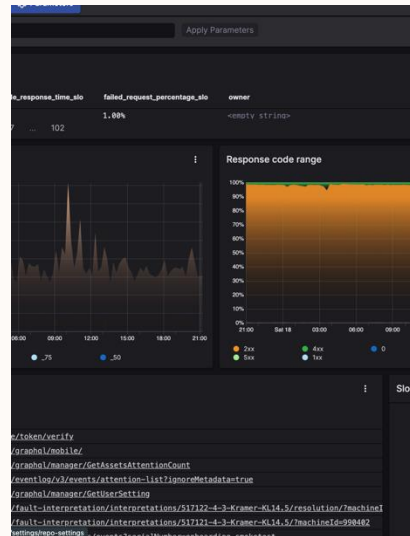
Security

You are responsible for ensuring your components are patched, secure and well designed from a security perspective



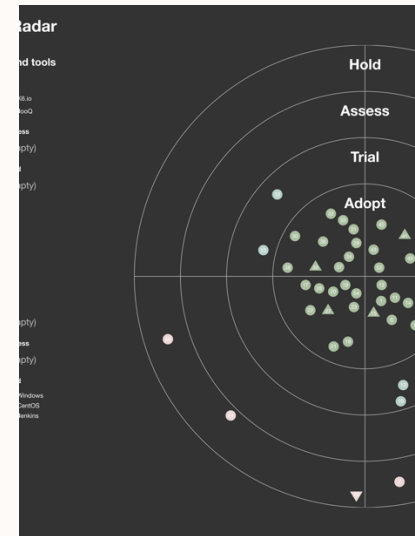
Availability

You are responsible for ensuring your component is up, fast and are also the one that will fix it if it breaks



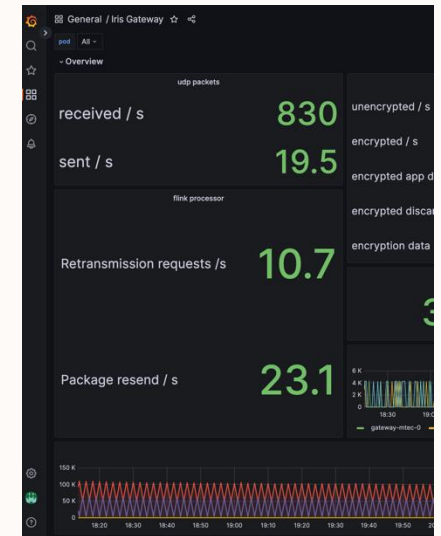
Maintainability

You are responsible for ensuring that you make technical decisions that favor maintainability.

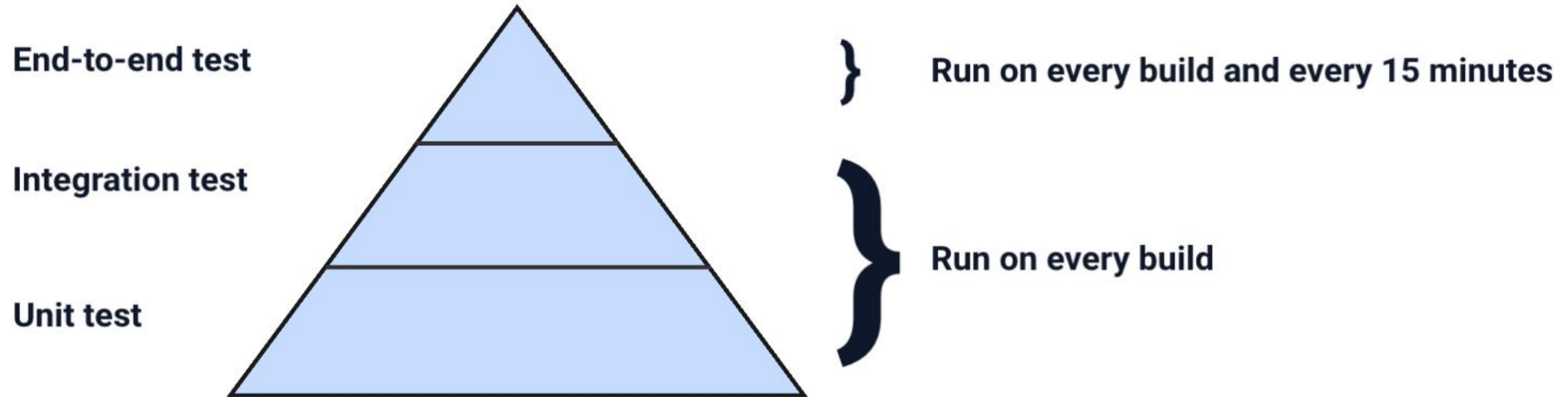


Observability

You are responsible for ensuring that your component can be monitored, has the right alerts, dashboards and logging.



Testing



Component ownership

Services 1071

Detected Services 418

+ New Service

Service List

Dependency Graph

Deploys

Service List 1071

Filter

Level		Name	Owner	Tier	Last Deploy	Tags	Tools
Silver	+	manager TypeScript	Team IRIS Foundation	Tier 1	41 minutes ago	app : manager + 9 more	
Silver	+	graphql-internal TypeScript GraphQL	Team IRIS Foundation	Tier 1	5 hours ago	app : graphql-internal + 9 more	
Silver	+	graphql-public TypeScript GraphQL	Team IRIS Foundation	Tier 1	5 hours ago	app : graphql-public + 9 more	
Silver	+	graphql-public-viewer TypeScript GraphQL	Team IRIS Foundation	Tier 2	5 hours ago	app : graphql-public-viewer + 9 more	
Silver	+	graphqlv2-manager TypeScript GraphQL	Team IRIS Foundation	Tier 1	5 hours ago	app : graphqlv2-manager + 9 more	
Non compliant	+	graphqlv2-report TypeScript GraphQL	Team Reboot		5 hours ago	app : graphqlv2-report + 9 more	
Silver	+	graphqlv3-mobile TypeScript GraphQL	Team IRIS Foundation	Tier 1	5 hours ago	app : graphqlv3-mobile + 9 more	
Silver	+	iris-app-sdk-server TypeScript NodeJS	Team IRIS Foundation	Tier 1	5 hours ago	app : iris-app-sdk-server + 9 more	
Silver	+	storybook TypeScript Storybook	Team IRIS Foundation	Tier 2	5 hours ago	app : storybook + 9 more	

Opslevel component view

Services / asset-api

asset-api

Silver

Contains asset data in API format

ActionsDownloadSettings

On CallNobody

Summary

Operations

Dependencies

API Docs

Tech Docs

Campaigns

Maturity Report

Service Overview

Domain: IRIS

System: IRIS-APIs

Product: IRIS

Owner: Team Kin

Language: Java

Framework: SpringBoot

Tier: Tier 2

Lifecycle Stage: Generally Available

Repositories

core-api-asset

Tags

app : asset-api

app.kubernetes.io/name : asset-api

app.kubernetes.io/version : 3.2.197

argocd.argoproj.io/instance : asset-api

buildconfiguration : API_v3

humio-parser : spring-boot

imported : kubectl-opslevel

owner : team_kin

type : api

Aliases

asset-api

Trackunit_core-api-asset

github.com:Trackunit/core-api-asset

k8s:asset-api-default

Notes

Add a quick reminder, troubleshooting tips, or any extra info that's important for this service.

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Opslevel component view

Services / asset-api

asset-api

Silver

Contains asset data in API format

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On CallNobody

Summary

Operations

Dependencies

API Docs

Tech Docs

Campaigns

Maturity Report

Environment: All

Tools

Edit

Sort By: Name

ApiDashboard-dev
Metrics | dev

ApiDashboard-prod
Metrics | prod

ApiDashboard-stage
Metrics | stage

FalconLogscale
Logging | dev

FalconLogscale
Logging | prod

FalconLogscale
Logging | stage

ResourceUsage-dev
Metrics | dev

ResourceUsage-prod
Metrics | prod

ResourceUsage-stage
Metrics | stage

Deploys

1W2W1M2024-03-27 21:00 ~ 2024-04-10 21:00

Today at 1:11 PM by CircleCI
Deployed asset-api version 3.2.197 to prod b92d7c

Show Details

Today at 11:24 AM by CircleCI
Deployed asset-api version 3.2.197 to dev b92d7c

Show Details

Today at 11:24 AM by CircleCI
Deployed asset-api version 3.2.197 to stage b92d7c

Show Details

Yesterday at 11:19 AM by Mateusz Kaczynski
Deployed asset-api version 3.2.197-SNAPSHOT-KPB_3580-1712654231 to dev 48ba8c

Show Details

April 8, 2024 at 10:32 PM by Mateusz Kaczynski

Opslevel driving component compliance

Services / asset-api

asset-api

Silver

Contains asset data in API format

ActionsDownloadSettings

On CallNobody

SummaryOperationsDependenciesAPI DocsTech DocsCampaignsMaturity Report

Service Maturity

Current Level

Silver

Services in this level satisfy important and critical checks. This is considered healthy.

Rubric

All Categories

Service Ownership

Security

Reliability

Performance

Scalability

Observability

Infrastructure

Quality

Cost control

Scorecards

All Scorecards

Flink

IRIS API golden path

Checks

Non compliant

Bronze

Silver

Gold

Total Checks Passing: 24 / 27 (88%)

> Bronze 13

> Silver 9

> Gold 5

Service has had service architecture review least 6 months

Required

Rubric | Service Ownership

Update

Error: Check Expired: No passing results were found for this check between May 30, 2023 19:16 and November 30, 2023 19:16 (UTC).

This manual checks is used to confirm that a recent (6 months) service architecture review has been carried out.

Next Update Due: May 30th, 2024 at 21:16:09 +02:00
Last Updated: December 1st, 2023 at 05:25:09 +01:00

Reviewed service dependency and updated

Required

Rubric | Observability

Update

This check has not been manually evaluated.



Maturity

Domain

All

System

All

Tier

All

Lifecycle

All

Saved Filter

Select a saved filter

Team

All

Include Subteams

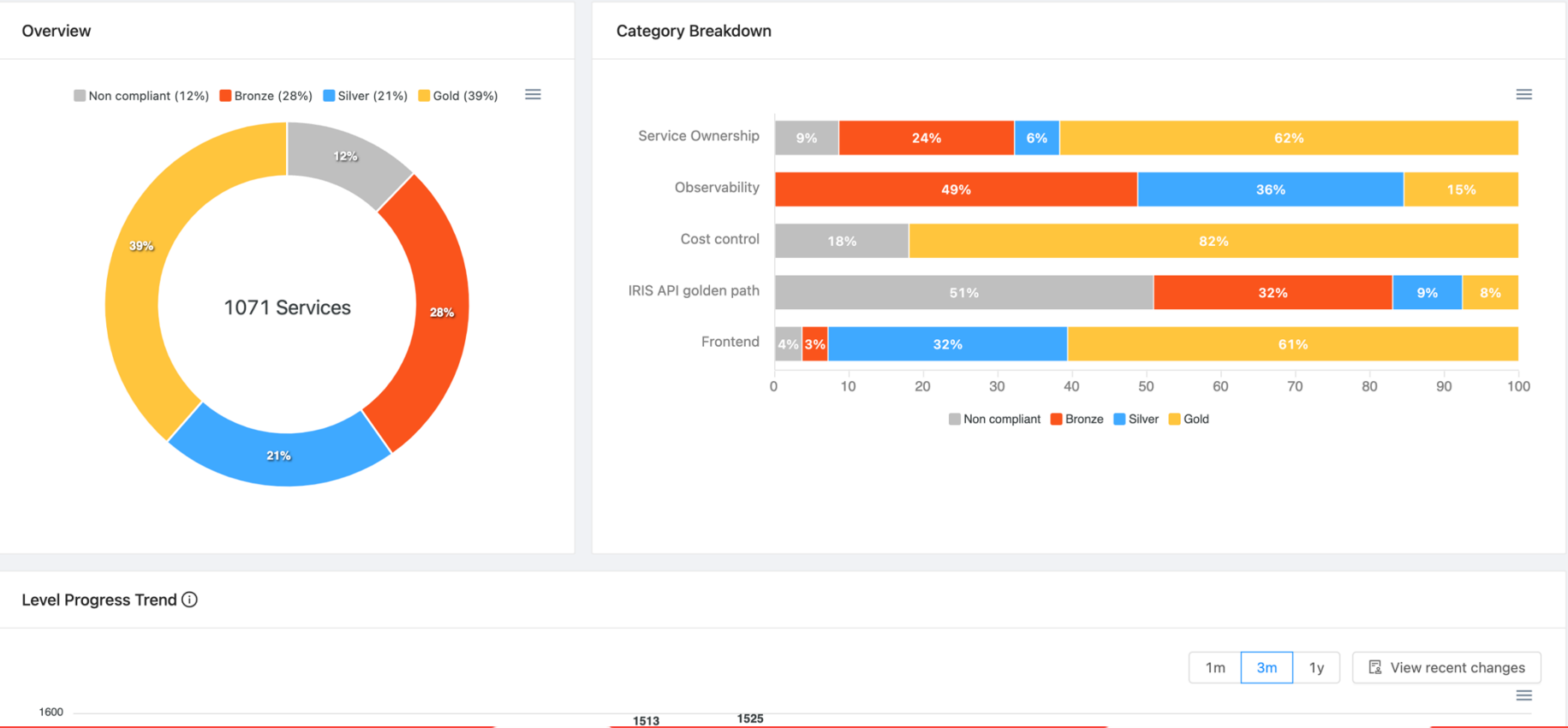
☒

Appears In

All

Levels

All



Level Progress Trend

1m 3m 1y

View recent changes

Driving operational maturity

Checks enforced in the pipeline around topics such as

- Service ownership
- Observability
- Cost control
- Iris API golden path
- Frontend

Campaigns

Campaigns / Secrets scanning

Secrets scanning

Owner: Team SRE

Start Date: March 5, 2024 12:00 PM (GMT+2)

Status

Default Filter: APIv3 builds

Target Date: April 5, 2024 10:00 AM (GMT+2)

Checks Passing 91%

Service 112 /

Overview

Campaign Checks

Project Brief

Trufflehog security scanning

In early january we introduced trufflehog security scanning as a replacement for Spectral ops. This was done as spectral ops was not flexible to fit our team structure and automated reporting, and came at a large cost. Due to this we decided to shift into trufflehog scanning, the check has now been in gold for 3 months, and we are preparing to move it into bronze. During this we will also set it to become blocking in the build pipeline.

When the check is moved to bronze it will take any failing components and downgrade it to non compliant. Any secrets should be stored in vault.

False positives.

You can remove false postivies like this:

Example:

```
yaml: password=notasecret #trufflehog:ignore
```

```
java: var password="notasecret"; //trufflehog:ignore
```

We will frequently review false postives marked, and this is not a way to ignore moving your secret to vault. We will also remove kubernetes secrets soon.

Timeline

On April the 5th we will move the check into bronze We will also set the build pipeline to fail PRs based on scan results.

Links

[Component expectations - security](#)

[Security at trakcunit](#)

Contacts

If you have any questions please reach out to:

[team iris SRE](#)

Sponsor

SBM - Søren Bruncke Mikkelsen, Responsible of security.

Trend

Trend Last Changed: Apr 5, 2024 at

Total Services

Passing Services

Find a service...

Status: All

Group by: Services

Service Table Last Changed: Apr 5, 2024 at

Status	Service Name	Secret scan	Tier	Owner	Contact
✓	access-management-	✓		Team Access control	#team-access-control



07

Our way to building a world class software platform with
the construction industry

Golden Path



Golden paths

- Opinionated and supported path to building software
- Multiple golden paths exists
- Governed by communities and teams
- By choosing to follow the golden path, you will get things out of the box, such as backup, monitoring or other





Postgres golden path

SEARCH

Home

Catalog

MY GROUPS

APIs

Docs

Explore

Tools

Create...

Quickstart

Catalog Graph

Tech Radar

TRACKUNIT TERRAFORM

Home

Methodology

Policies

Applications

AWS Node Termination Handler

Classic AWS Infrastructure

Databricks E2

File store

Firmware FTP

Okta

PostgreSQL

Golden path

Change data golden path

Secret Accessor

Transit Gateway

Twilio API keys

Vault

Terraform Modules

Postgres databases

This document describes our golden path for utilizing postgres databases, and how we are hosting them.

Following the golden path, means that feature teams can be sure to get support and help with maintenance, and will get things for given such as monitoring, backup and infrastructure network access required to efficiently work with the database.

The golden path sets out to build per microservice databases, that can be used in the context of a given service. Best practice is not to share a single database between multiple services, please ensure that as a developer you are not compromising the microservice architecture and creating hard couplings between services on a non service boundary layer (eg. by doing direct database integrations)

From a developers point of view

The following section contains the information you need from a developers point of view, to order, access and use a database.

Requesting a new database

We have chosen a hybrid cluster architecture where databases will be hosted either separately or in a general cluster depending on usage patterns. Databases might be moved as they grow.

To order a new database you should provide the following to the sre team:

Item	Description	Example
component name	The name of the API or system component you wish to acquire a new database for	tas

Table of contents

From a developers point of view

What will I receive

Multi schema support

Retrieving secrets from Vault

Connecting to the database

Responsibilities of the developer

What is taken care of for me

Technical specification and infrastructure overview

Aurora endpoints

Tiering

Environments

Options set on the clusters

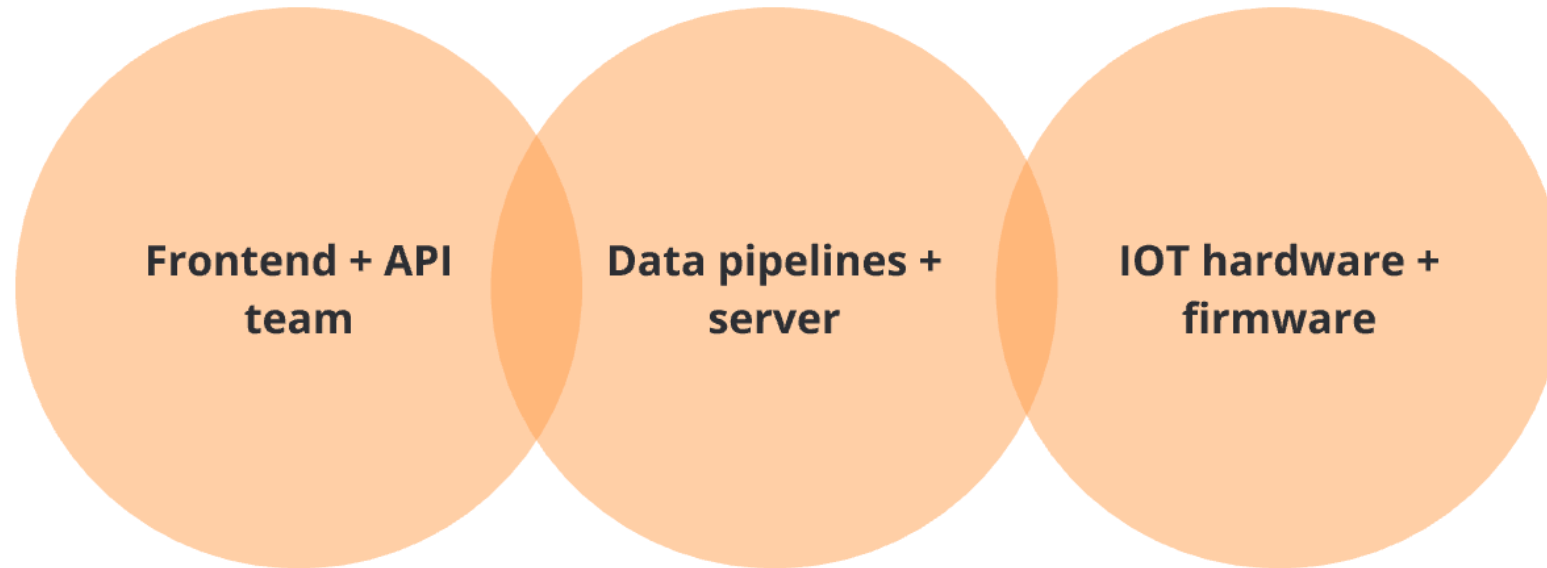


08

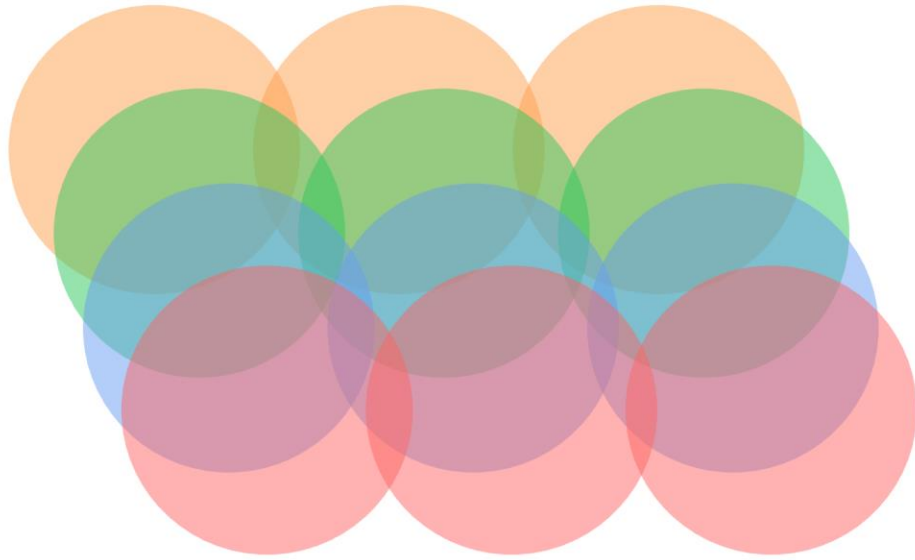
**Scaling an
engineering
organization**

2018

~ 25 people



Scaling to 200 people



🧑‍🤔 COVID-19



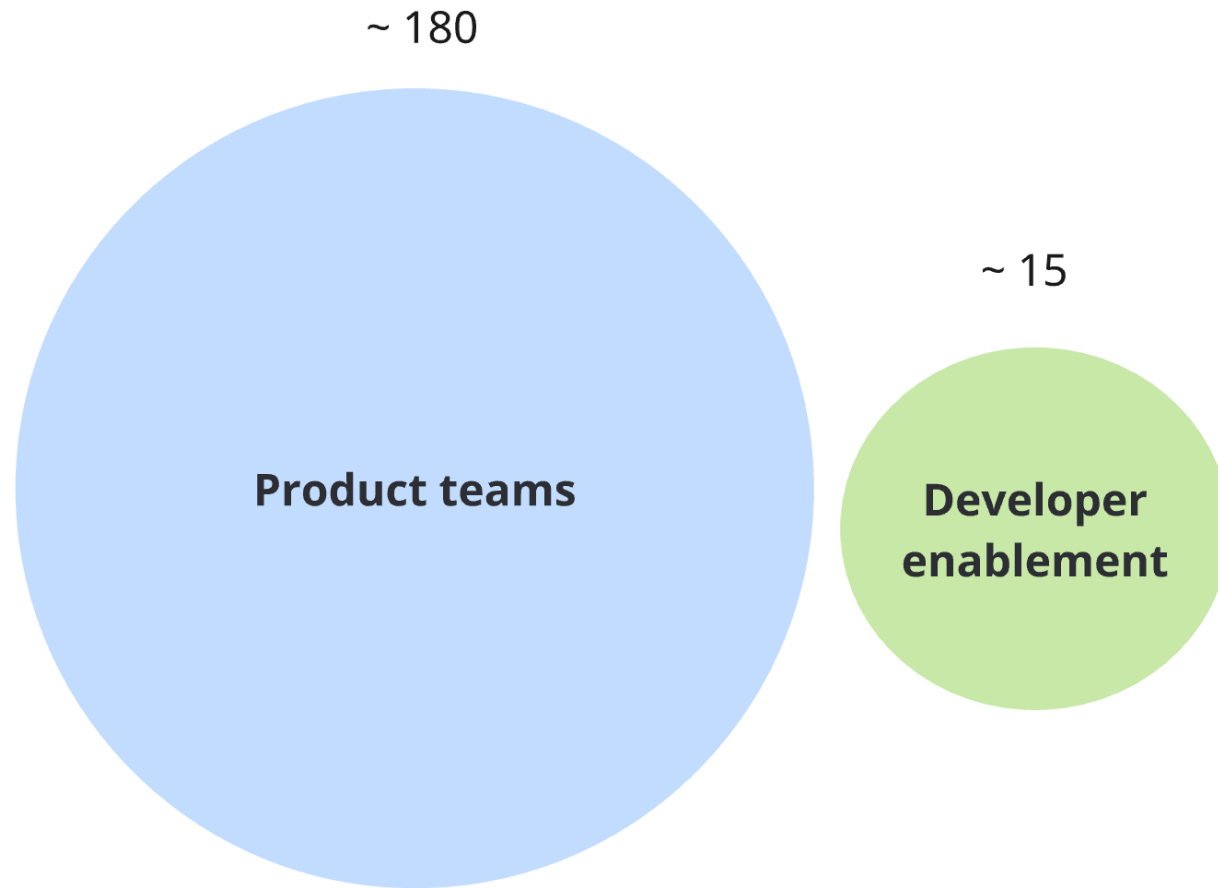
New challenges

- Multiple solutions to crosscutting concerns were made
- Larger code base that need to be maintained
- Reorganizing teams led to difficulty maintaining code
- Specialist knowledge required to scale (security, databases, servers, API design)
- Higher focus on security
- Management lacked insight into engineering org performance
- Higher focus on uptime and incident management
- 2018 alignment mechanisms did not work
- Production system configuration overview missing

What did we change

- Introduced developer enablement team handling crosscutting concerns
- Aligned technologies and tools used
- Introduced communities to facilitate knowledge sharing
- Built golden paths to support common development work
- Measure the performance of the engineering organization
- Introduced incident process to ensure consistent incident handling
- Introduced team ownership
- Adopt declarative approach to running software and use Gitops

Team constellation - 2025



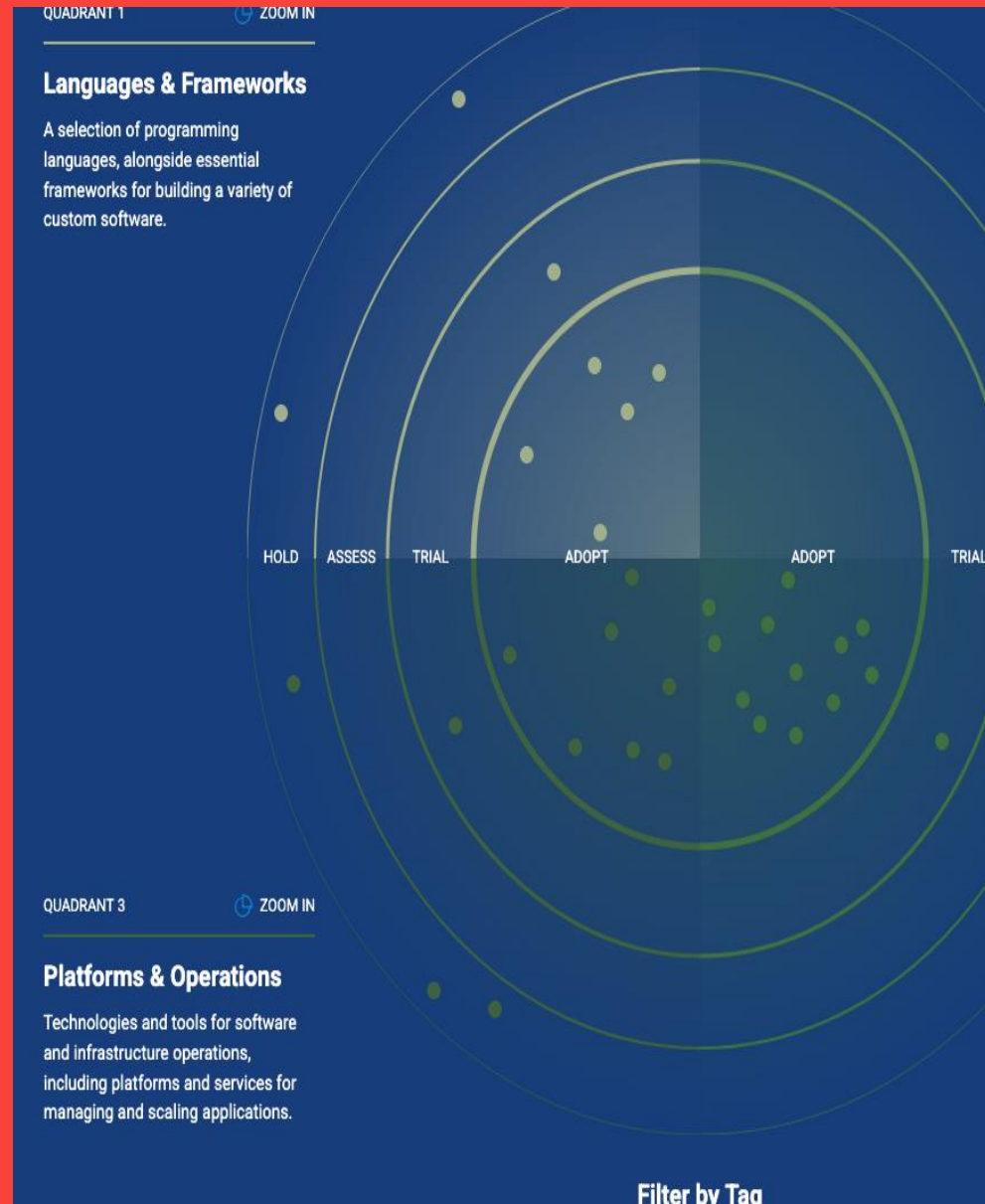
Developer enablement team

Responsibilities

- Servers
- Network
- Databases – traditional and NoSQL
- Monitoring stack – application logs, metrics and traces
- Backup
- Kubernetes

Provides teams a simple interface to run applications

Product teams need only a minimal understanding about the above



Tech radar

- Familiar tech
- Focus on fewer technologies





Creating alignment

Communities

Monthly meeting + slack channel

Spring Boot Community Knowledge Sharing
November 5th 14-15 CET

Hey team!

Our next **Spring Boot Community** knowledge sharing

✨ Already confirmed:

- Best practices on kafka + running APIs locally (1h)
- "How I got Cursor and SpringBoot succeed - what I learned"
- "How to break stuff with Spring Boot 4" (20m)

Got a tip, trick, or idea you'd like to share? Big or small. Drop your name here or ping me directly 🙋

Let's make this session packed with learning and planning

Written documentation



Engineering Handbook

Design Docs

What Is the Engineering Handbook

Trackunit Engineering >

Iris >

Iris Architecture >

Domain Modelling >

Backend Development >

Plan for async work

 Aalborg, Denmark CEST (UTC +2)	Thu, 2 Oct 2025	20:00	😞
 London, Canada EDT (UTC -4) 6 hour(s) behind	Thu, 2 Oct 2025	14:00	😊
 Wroclaw, Poland CEST (UTC +2) same as Aalborg	Thu, 2 Oct 2025	20:00	😞

09:00, Wed, 1 Oct 2025

Fri, 3 Oct 2025, 00:00

<< Current Time >>

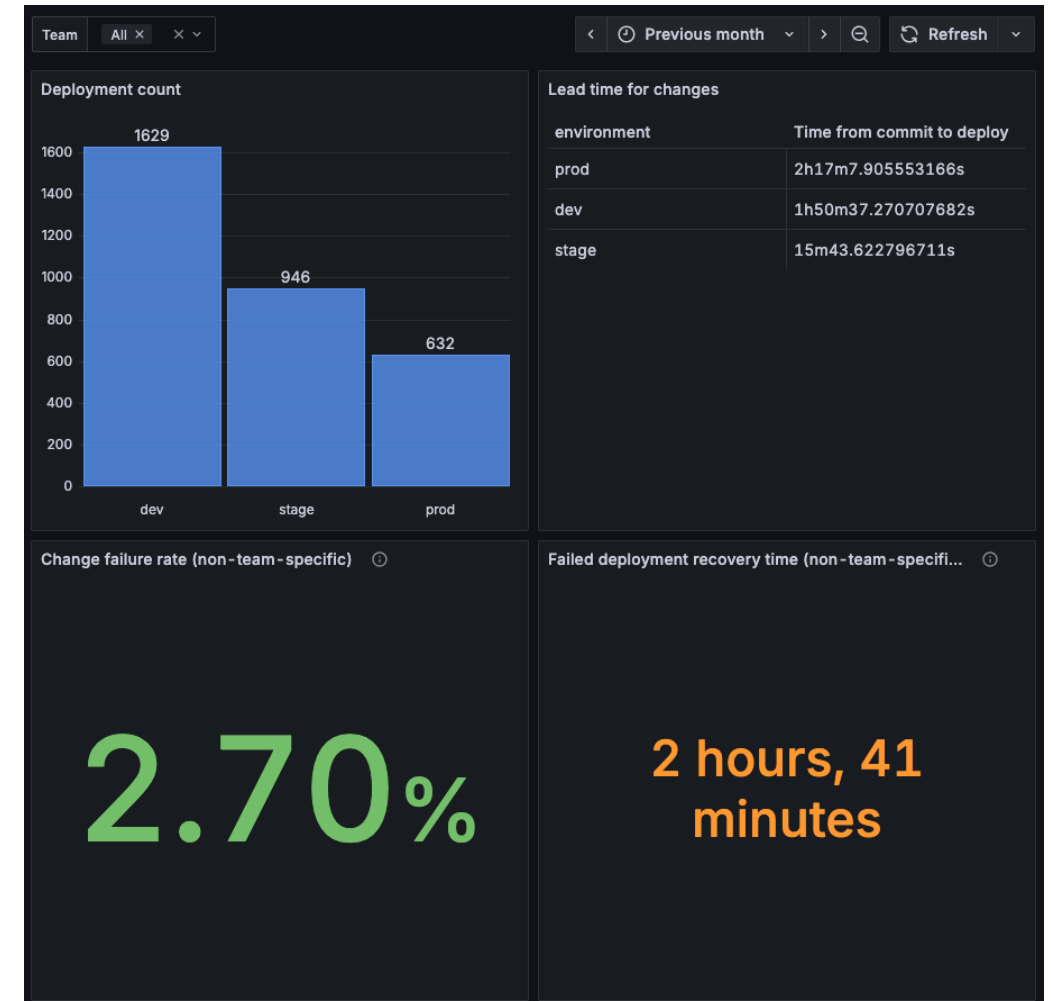
Organization performance

What to measure? Lines of code written?

DevOps Research and Assessment (DORA)

suggests measuring

- How long it takes to make a new release
- How often you deliver a new release to customers
- How often a delivery is faulty (in %)
- How long it takes to fix a faulty release



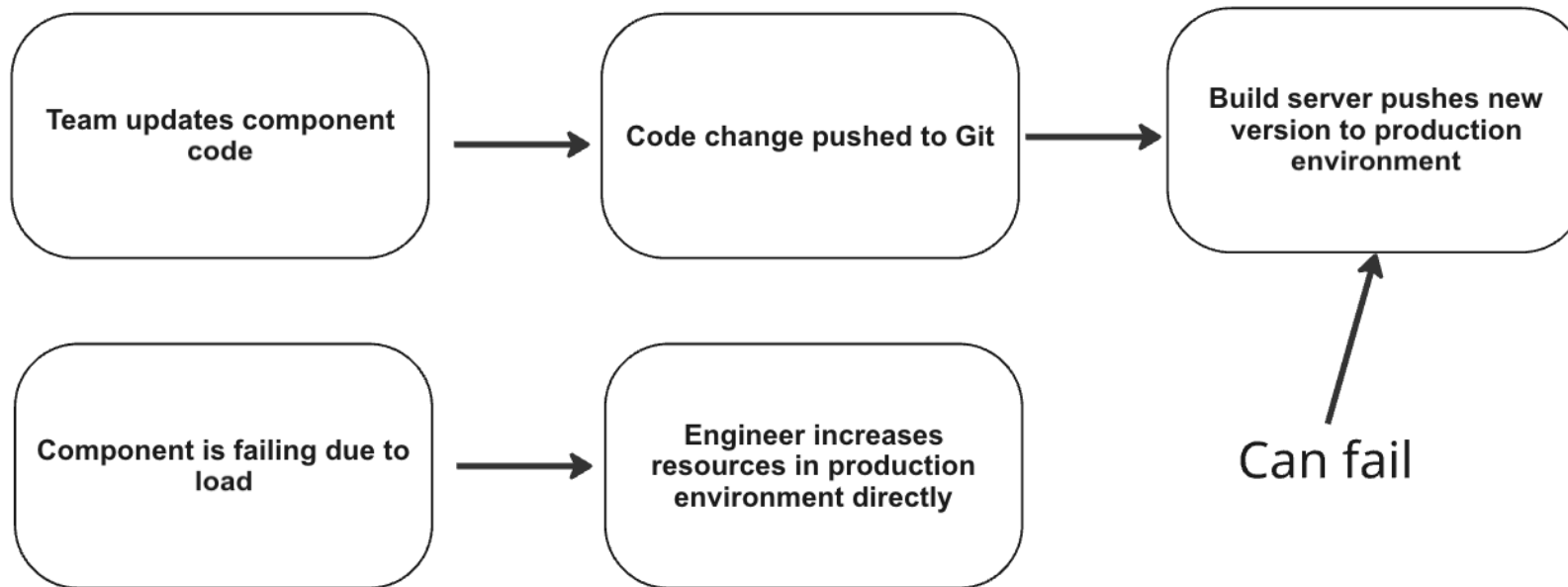
Declarative configuration

“Start two instances of telematics-device-api” (action)

vs

"Two instances of telematics-device-api are running" (desired state)

Before Gitops



After Gitops

